

ODESSA-SCHLEMEYER FIELD, TX, USA

WMO: 722648

Lat: 31.921N

Lon: 102.387W

Elev: 915

StdP: 90.81

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 03031

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.0	-3.7	-17.4	0.9	7.2	-15.5	1.1	8.9	12.8	11.5	11.8	10.0	3.7	40	0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	38.2	18.8	37.0	19.2	35.9	19.2	22.4	30.7	21.9	30.3	21.5	29.8	5.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.5	17.0	24.3	19.9	16.3	24.2	19.2	15.6	24.3	70.3	30.8	68.5	30.2	66.9	29.7	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.5	10.2	9.0	DB	-8.7	40.9	2.2	1.6	-10.3	42.0	-11.6	42.9	-12.8	43.8	-14.4	44.9	(4)
			WB	-10.0	23.4	2.0	0.8	-11.4	23.9	-12.6	24.4	-13.7	24.8	-15.1	25.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	18.6	7.6	9.8	14.8	19.2	23.4	28.0	28.3	28.2	24.2	18.9	12.5	7.9	(6)
		DBStd	8.77	4.95	5.82	5.20	4.58	4.36	2.77	2.46	2.44	3.29	4.72	5.05	4.81	(7)
		HDD10.0	331	106	68	18	2	1	0	0	0	4	33	98		(8)
		HDD18.3	1322	333	242	132	44	11	0	0	2	50	182	325		(9)
		CDD10.0	3471	32	62	166	277	417	539	568	564	426	280	108	32	(10)
		CDD18.3	1419	0	2	21	69	169	289	310	306	178	68	7	0	(11)
		CDH23.3	17006	10	66	312	865	1994	3626	3847	3766	1768	652	86	13	(12)
(13)		CDH26.7	8644	1	12	83	342	1010	2027	2100	2046	786	225	14	0	(13)
(14)	Wind	WSAvg	4.7	4.2	4.6	5.1	5.5	5.4	4.8	4.4	4.3	4.4	4.2	4.2	4.2	(14)
(15)	Precipitation	PrecAvg	327	16	18	17	17	36	34	42	46	36	33	17	13	(15)
(16)		PrecMax	588	44	55	52	58	181	79	158	117	114	103	124	69	(16)
(17)		PrecMin	143	1	0	2	0	0	1	0	1	0	0	0	0	(17)
(18)		PrecStd	114	14	16	13	18	43	23	40	30	30	32	26	17	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	24.8	28.6	31.4	34.1	38.5	40.6	39.1	38.8	36.4	33.0	28.7	25.3		(19)
(20)		0.4% MCWB	10.6	12.6	13.1	14.1	16.4	17.7	19.8	19.8	19.2	16.9	13.8	11.0		(20)
(21)		2% DB	22.2	25.2	29.0	32.5	36.3	38.6	37.4	37.3	34.3	31.2	25.9	22.1		(21)
(22)		2% MCWB	9.8	10.7	12.8	13.9	16.0	18.0	19.9	20.0	19.0	17.1	13.8	10.1		(22)
(23)		5% DB	19.3	22.9	27.0	30.6	34.2	37.0	36.2	36.2	32.8	29.3	23.6	19.0		(23)
(24)		5% MCWB	8.5	10.3	12.2	13.8	16.0	18.8	19.9	20.1	18.8	16.9	12.7	9.1		(24)
(25)		10% DB	16.8	20.2	24.7	28.4	32.4	35.1	34.8	34.7	31.2	27.3	21.4	16.5		(25)
(26)		10% MCWB	8.0	9.2	11.8	13.9	16.2	19.4	19.9	20.1	18.9	16.2	11.9	8.4		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	14.0	15.1	17.7	19.4	21.6	23.0	22.9	22.3	21.6	18.1	14.4			(27)
(28)		0.4% MCDB	17.5	20.4	23.3	26.3	29.1	31.7	30.7	31.3	29.8	27.5	21.6	17.9		(28)
(29)		2% WB	11.7	13.5	16.5	18.4	20.8	22.2	22.2	22.3	21.6	20.4	16.2	12.2		(29)
(30)		2% MCDB	17.1	19.6	21.4	24.4	27.5	30.6	30.4	31.1	28.8	25.7	21.0	17.7		(30)
(31)		5% WB	10.0	11.9	15.4	17.4	20.1	21.7	21.7	21.8	21.0	19.6	15.0	10.6		(31)
(32)		5% MCDB	17.8	20.0	21.0	23.7	26.7	30.0	30.1	30.7	27.9	24.6	19.9	17.3		(32)
(33)		10% WB	8.4	10.4	14.1	16.3	19.4	21.1	21.3	21.3	20.5	18.6	13.7	9.0		(33)
(34)		10% MCDB	16.4	18.9	20.8	22.8	26.1	29.6	29.7	30.0	27.1	23.8	19.3	16.4		(34)
(35)	Mean Daily Temperature Range	MDBR	13.8	14.2	14.7	14.8	13.7	13.2	11.8	11.9	11.6	12.9	13.4	13.3		(35)
(36)		5% DB MCDBR	17.8	18.1	17.2	16.7	16.2	15.1	13.3	13.3	13.7	14.7	15.8	17.4		(36)
(37)		5% DB MCWBR	8.6	8.1	6.9	6.7	6.1	4.4	3.5	3.3	4.1	5.6	6.8	8.6		(37)
(38)		5% WB MCDBR	15.3	15.4	13.7	13.7	13.6	13.0	11.6	11.7	10.7	11.1	12.2	14.6		(38)
(39)		5% WB MCWBR	8.3	8.0	6.7	6.7	6.2	4.2	3.3	3.3	3.6	5.2	6.6	8.2		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.263	0.273	0.292	0.326	0.346	0.375	0.395	0.370	0.350	0.311	0.284	0.268		(40)
(41)		taud	2.582	2.555	2.516	2.408	2.417	2.384	2.371	2.416	2.440	2.540	2.546	2.566		(41)
(42)		Ebn at Noon	980	997	994	963	940	907	887	909	917	939	943	954		(42)
(43)		Edh at Noon	79	90	101	118	118	121	122	115	108	91	82	77		(43)
(44)	All-Sky Solar Radiation	RadAvg	3.49	4.33	5.60	6.84	7.27	7.63	7.24	6.71	5.61	4.60	3.66	3.12		(44)
(45)		RadStd	0.33	0.44	0.40	0.42	0.48	0.35	0.39	0.29	0.38	0.49	0.38	0.28		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page